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貨架選擇最佳化於自動化揀貨系統

**Optimization of Picking Pod Selection in Robotic
Mobile Fulfillment System**

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摘要

電子商務的爆炸性增長，受全球疫情的推動，已經引發了消費者行為和倉庫運營的範式轉變。本論文針對電子商務倉庫中訂單履行優化的緊迫挑戰，通過改進智動化揀貨系統（RMFS）來解決這一問題。研究特別針對開發高效的貨架選擇算法，以提高 RMFS 在管理不斷增長的訂單量和複雜性方面的整體性能。本研究引入了一種新穎的兩步貨架選擇程序。這種創新方法集成了匈牙利算法，以解決機器人和貨架之間複雜的多對多分配問題。研究的主要目標有三個：最大化訂單揀選吞吐量、最小化貨架移動、以及優化訂單整合以提高每次貨架訪問的效率。該方法通過廣泛的模擬來嚴格評估所提算法在各種運營場景中的有效性。這些模擬設計用來複製現實世界的倉庫條件，考慮了訂單量波動、產品多樣性和倉庫佈局配置等變量。新方法的性能與現有的貨架選擇策略進行了仔細的對標，評估了每次貨架訪問的揀選物品數量和整體系統吞吐量等關鍵性能指標。結果表明，所提的兩步貨架選擇算法在運營效率方面相對於傳統方法有顯著提高。尤其是集成了匈牙利算法的機器人-貨架分配顯示出在資源分配優化和最小化閒置時間方面具有特殊的潛力。此外，研究顯示，增強的訂單整合策略大幅增加了每次貨架訪問的揀選物品數量，從而顯著提高了整體系統的生產力。

關鍵字：RMFS、揀貨、訂單吞吐量、貨架選取

ABSTRACT

The exponential growth of e-commerce, catalyzed by the global pandemic, has precipitated a paradigm shift in consumer behavior and warehouse operations. This thesis addresses the pressing challenge of optimizing order fulfillment in e-commerce warehouses through the enhancement of Robotic Mobile Fulfillment Systems (RMFS). The research specifically targets the development of an efficient pod selection algorithm to improve the overall performance of RMFS in managing escalating order volumes and complexity.

The study introduces a novel two-step pod selection procedure. This innovative approach integrates the Hungarian algorithm to resolve the intricate many-to-many assignment problem between robots and pods. The primary objectives of this research are threefold: maximizing order-picking throughput rates, minimizing pod movements, and optimizing the consolidation of orders to enhance efficiency during each pod visit. The methodology employs extensive simulations to rigorously assess the efficacy of the proposed algorithm across a diverse array of operational scenarios. These simulations are designed to replicate real-world warehouse conditions, accounting for variables such as order volume fluctuations, product diversity, and warehouse layout configurations. The performance of the new approach is meticulously benchmarked against existing pod selection strategies, evaluating key performance indicators including the number of items picked per pod visit and overall system throughput efficiency. Results indicate that the proposed two-step pod selection algorithm demonstrates significant improvements in operational efficiency compared to conventional methods. The integration of the Hungarian algorithm for robot-pod assignment shows particular promise in optimizing resource allocation and minimizing idle time. Furthermore, the study reveals that the enhanced order consolidation strategy substantially increases the number of items picked per pod visit, leading to a marked improvement in overall system productivity.

Keywords: RMFS, Pod Selection, Order Picking, Throughput efficiency

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CHAPTER 1

INTRODUCTION

1.1 Background

The global pandemic has rapidly transformed electronic commerce, significantly altering consumer behavior. This shift has introduced unparalleled convenience and flexibility, allowing shoppers to browse numerous products, compare prices, read reviews, and make purchases from the comfort of their homes or on the go with just a few clicks. This ease of access has led to a rise in impulse buying.

E-commerce offers several advantages over traditional retail models [1]. For consumers, it removes geographical barriers, providing access to products from around the world. In warehouses, the range of product combinations is vast, with unit sizes often being very small and still largely reliant on manual picking operations. As e-commerce continues to grow, order volumes increase annually, necessitating more robust and efficient management systems. These systems are essential for improving order fulfillment, enhancing competitiveness, and boosting customer satisfaction.

The rapid growth of e-commerce has underscored the need for more efficient and adaptable warehouse management systems. As order volumes and complexity increase, traditional manual picking operations are struggling to keep pace. In response, many e-commerce businesses are turning to advanced technologies such as Robotic Mobile Fulfillment Systems (RMFS) to optimize their operations. These systems have the potential to significantly reduce labor costs while greatly enhancing order fulfillment accuracy and speed. However, the effectiveness of RMFS depends on optimizing various operational aspects. Among these, the pod selection procedure emerges as a crucial element in enhancing overall system performance. As the e-commerce landscape continues to evolve, addressing this key component becomes increasingly vital for maintaining competitiveness and meeting customer expectations.

Optimizing the pod selection procedure for e-commerce facilities is the main objective of this research. In order to achieve this main goal, this study develops an efficient algorithm for pod selection and implements a two-step procedure that considers pods currently in use and those in storage. The proposed method updates the appropriate order set based on the availability of items in active pods. By utilizing the Hungarian algorithm to solve the many-to-many assignment

problem, this study aims to optimize the assignment of robots to pods, considering retrieval and idle robots to ensure maximum efficiency. Top priorities will be enhancing order-picking throughput rates and developing strategies to fulfill more orders within a specified timeframe while maintaining a balance between efficiency and speed. This study reduces the number of pod visits to pick stations, minimizes unnecessary pod movements, and develops strategies to consolidate orders for maximum efficiency during each visit. This study aims to increase the number of items picked during each pod visit, optimize the products a worker can pick from one pod each time it arrives (pile-on), and implement order grouping algorithms to enhance efficiency. Ultimately, through extensive simulations, this study assesses the proposed pod selection algorithm's effectiveness in scenarios and compares it with other pod selection strategies to analyze and evaluate its performance. By accomplishing these related objectives, this study aims to significantly advance the field of warehouse automation by enhancing RMFS efficiency, especially in the critical area of pod selection. This ultimately leads to improved warehousing operations, increased customer satisfaction through expedited order fulfillment, and enhanced competitiveness in e-commerce.

1.2 Problem Statement

The expansion of online retail has placed increasing pressure on warehouse to process a growing volume of order efficiently. To attract customers, some companies go as far as guaranteeing afternoon delivery for morning orders, which places pressure on warehouses to meet tight delivery deadlines.

The two primary functions in warehouses are replenishment and picking. These procedures work best when used together, and efficient replenishment strategies ensure that there are always enough products available for selection. The picking procedure is the primary issue that needs to be resolved because it is responsible for about half of warehouse operational expenses. The main goal of this study is to optimize pod (shelf) selection during picking, with a focus on utilizing autonomous mobile robots (AMRs) to enhance efficiency.

1.3 Research Objectives

This study aims to develop an algorithm for pod selection that can improve the efficiency of the order-picking process. The proposed algorithm implements a two-step procedure that considers pods currently in use and those in storage.

1.4 Scope and Limitations

This study focuses on robotic mobile fulfillment systems (RMFS) and e-commerce warehouse operations. Determining the scope of this investigation is crucial:

- The research aims to optimize internal warehouse procedures, specifically those related to order fulfillment.
- The suggested solutions are tested and validated using simulation-based approaches.
- The study acknowledges that RMFS encompass many functions, but just focuses on order picking.

Limitations:

- The analysis excludes external factors such as market developments and supply chain dynamics.
- The layout and configuration of the warehouse is fixed.
- This study only focuses on pod selection.
- There is no explicit modeling of the influence of seasonal variations in demand.
- Despite being grounded in real-world situations, simulations may not fully capture the complexities of authentic warehouse settings.

These well-defined parameters acknowledge the broader context of warehouse management while facilitating a focused investigation of the study's issues. The results must be understood within these limitations, bearing in mind that real-world application would require consideration of other aspects not covered in this investigation.

1.5 Thesis Structure

A comprehensive analysis of relevant literature is presented in Chapter 2. It examines the literature currently available on RMFS, pod selection techniques, and warehouse optimization algorithms. The theoretical underpinnings and the research gap that this study seeks to address are both identified in this chapter. A thorough explanation of the RMFS pod selection problem is provided in Chapter 3. It codifies the decision-making procedure, which includes robot-pod assignment using the Hungarian algorithm and a two-stage pod selection process. The mathematical formulation of the problem, which incorporates the goals of maximizing pick efficiency, minimizing pod visits, and optimizing throughput, is also presented in this chapter. The

presentation and analysis of numerical studies are covered in Chapter 4. The text explains the experimental methodology, simulation setup, and performance indicators that are used to assess the suggested approach. This chapter presents the findings, contrasting the performance of new concepts with current tactics and offering an analysis of their effectiveness in various operational contexts. The thesis is concluded in Chapter 5, which provides a summary of the major discoveries and contributions of the research. It considers how the findings may affect e-commerce warehouse operations and makes recommendations for future research on optimizing RMFS.

CHAPTER 2

LITERATURE REVIEW

2.1 Warehouse

Warehouses are essential to the supply chain because they act as central locations for managing and storing items before their delivery to the end customer. Distribution centers and fulfillment centers are the two main types of warehouses. Distribution centers primarily handle the bulk transfer of goods to different retailers, whereas fulfillment centers specialize in processing and shipping individual customer orders, playing a crucial role in the e-commerce sector. This research will focus on fulfillment centers, considering their significance in the modern retail landscape.

The typical flow of goods in a warehouse involves several key processes. Initially, goods are received and checked for accuracy and quality. Following the reception, items are labeled and stored systematically to ensure easy retrieval. The order-picking process then begins, during which products are selected to fulfill customer orders. Items are gathered, sorted, and packaged for shipping once they are selected. Each of these stages is essential to maintaining accuracy and efficiency in warehouse operations.

Warehouse systems [2] can be broadly categorized into two main groups: manual and automatic (see Figure. 1). For tasks such as picking and sorting, manual systems primarily rely on human labor, while automation systems utilize machinery to carry out similar activities. There are also some new methods designed to enhance precision and speed further. These include multi-shuttle systems, which use numerous shuttles for quick and flexible storage and retrieval; automated and robotized systems, which employ robots for activities like order picking; and smart systems, which integrate advanced technologies such as AI and IoT to optimize warehouse operations.

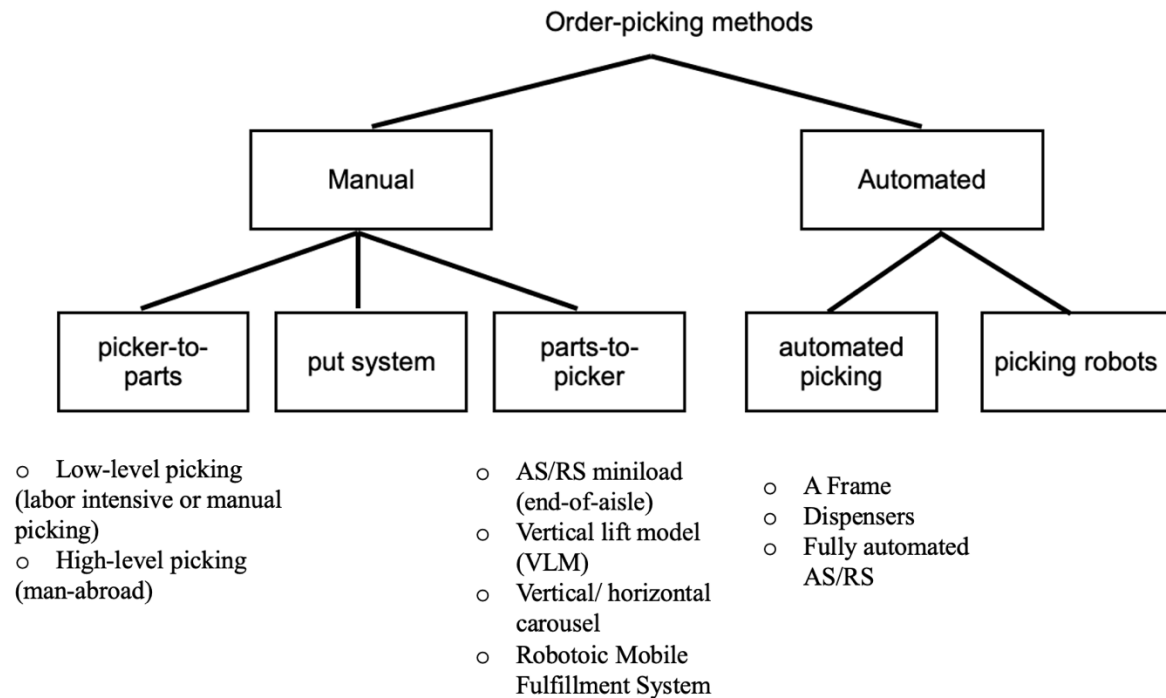


Figure 2.1 Order Picking Methods

2.2 Warehouse Fulfillment System

The development of industrial mobile robots began in the 1950s with simple wire-guided vehicles. Twenty years later, the first Automated Guided Vehicles (AGVs) were introduced in warehouses, following fixed paths using magnetic strips or wires embedded in the floor. A significant breakthrough occurred in 2003 with the introduction of the Kiva system. Kiva's robots can lift and move entire shelving units, delivering them to stationary human workers, significantly reducing walking time and improving efficiency. Warehouse automation underwent a significant transformation in 2012 when Amazon acquired Kiva Systems and rebranded it as Amazon Robotics.

This acquisition spurred the emergence of numerous startups aiming to fill the market gap. Companies such as Fetch Robotics, Grey Orange, and inVia Robotics have developed more flexible systems that utilize advanced sensors and artificial intelligence to navigate dynamically without fixed paths. E-commerce has rapidly increased as a result of the COVID-19 pandemic, putting significant strain on fulfillment centers and accelerating the adoption of robotic systems to manage the increased quantities.

Modern autonomous mobile robots (AMRs) now perform a variety of tasks beyond simply moving shelves. They further enhance the effectiveness and versatility of robotic systems in fulfillment operations by integrating with warehouse management systems (WMS) for real-time

optimization.

Two main methods are employed in warehouse operations: parts-to-picker and picker-to-parts [3, 4]. In the picker-to-parts method, pickers travel to the storage area to locate and retrieve target items. This method can be time-consuming and physically demanding as it involves extensive searching and walking, which can lead to picker fatigue.

In contrast, the parts-to-picker method is exemplified by the Robotic Mobile Fulfillment System (RMFS). In an RMFS, robots such as AGVs or AMRs bring pods containing the target Stock Keeping Units (SKUs) to the picking station. Here, pickers guided by the WMS fulfill orders by selecting the required items from the pods. This method significantly reduces the walking time for pickers and enhances the efficiency of the order-picking process see in Figure 2.2.

The RMFS enhances the operational flexibility of the system, making it adaptable to various warehouse environments and capable of efficiently handling different types of products. Warehouses can enhance their ability to meet the demands of the rapidly growing e-commerce market by boosting productivity, reducing labor costs, and improving order fulfillment accuracy through the adoption of robotic technologies and efficient management strategies.

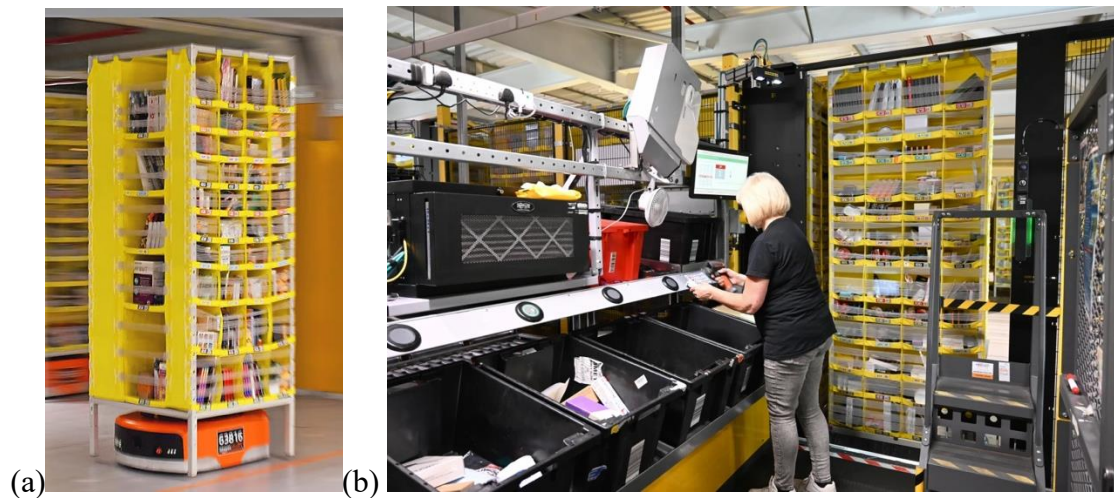


Figure 2.2 (a) A Pod Driven by A Mobile Robot in RMFS. (b) Human-robot Collaborative Order Picking in RMFS [5-7]

2.3 Order picking

Order picking is a critical component of warehouse operations, accounting for nearly half of the total operating costs. This process is the first step in fulfilling customer orders, and its efficiency directly impacts the subsequent stages of order processing, including packing and shipping. Ensuring smooth and efficient order picking is essential for timely order fulfillment and overall customer satisfaction.

There are several order picking strategies [2] employed to enhance efficiency and accuracy.

- **Batch-Picking:** Involves picking multiple orders simultaneously to reduce the number of trips made by pickers to the storage area.
- **Zoning:** Organizes the warehouse into different zones based on criteria such as product size, temperature requirements, or safety considerations. Pickers are assigned to specific zones, which enhances specialization and reduces travel time.
- **Sequential Zoning (Pick-and-Pass):** Items are picked sequentially across different zones. Pickers in each zone add the required items to the order before passing it to the next zone.
- **Parallel Zoning (Pick-and-Merge):** Multiple pickers work simultaneously in different zones, and their picked items are subsequently combined to fulfill the order.
- **Sorting Methods:** Include strategies like Sort-While-Pick, where items are sorted as they are picked, and Pick-and-Sort, where items are picked first and then sorted.

Order picking is particularly significant because it is labor-intensive, requiring substantial human effort and time. By improving management practices and optimizing task assignments, warehouses can significantly reduce time consumption and enhance overall efficiency. Effective order picking not only reduces operational costs but also ensures that customer orders are processed accurately and delivered on time.

As mentioned earlier, the manual system of picker-to-parts and parts-to-picker methods, particularly in systems like the Robotic Mobile Fulfillment System (RMFS), plays a crucial role in improving order-picking efficiency. These methods help minimize the time spent by pickers traveling and streamline the picking process, thereby enhancing productivity and reducing labor fatigue.

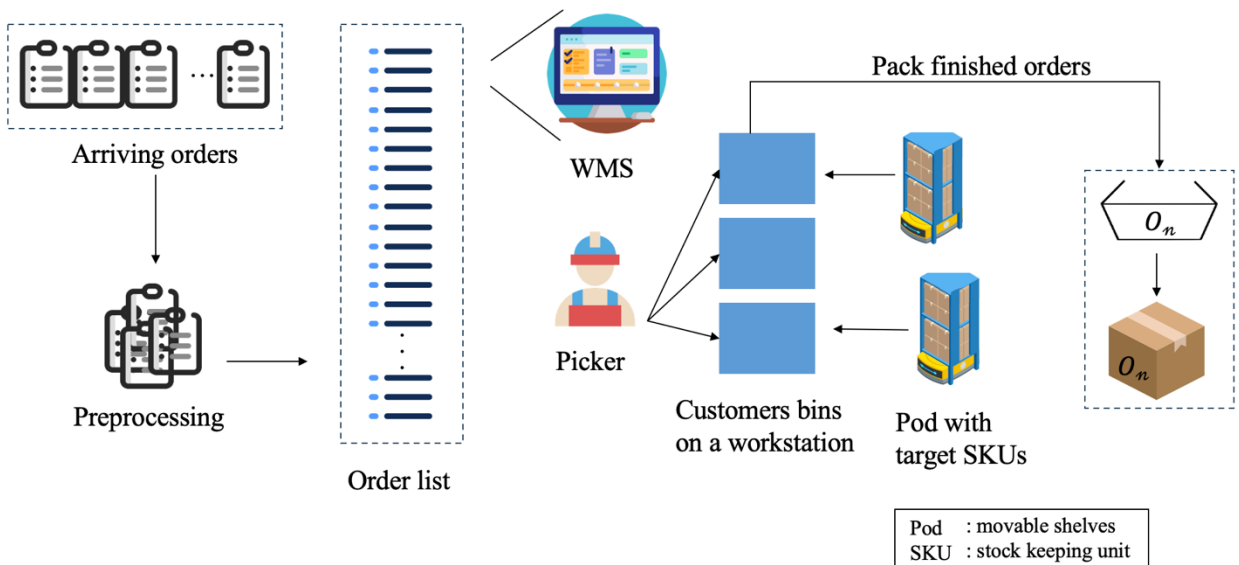


Figure 2.3 Elementary Picking Workflow at Each Workstation [8]

2.4 Robotic Mobile Fulfillment System

The focus on Robotic Mobile Fulfillment Systems (RMFS) is particularly relevant in Taiwan due to the country's limited land area and high population density. During the transition to semi-automation, RMFS offers a practical solution by introducing Autonomous Mobile Robots (AMRs) to assist human pickers. This approach enhances efficiency without the need for extensive machinery or large-scale restructuring of warehouse layouts. Within a constrained budget, RMFS can significantly improve order throughput, making it a valuable area of study.

System Analysis, design optimization, and operations planning and control are the three primary categories into which the literature on RMFS is divided [9]. In system analysis, [10] developed a closed-queueing network model to estimate order throughput time, while [11] expanded on this by incorporating a Semi-open queueing network (SOQN) to account for order cycle time for both single-line and multi-line orders. In design optimization, [11] found that the maximum throughput capacity in the storage zone is generally unaffected by the length-to-width ratio but is significantly influenced by the placement of workstations. [12] optimized the system's shape using an SOQN. In operations planning and control, [10] demonstrated that the closest-open location pod storage strategy, while less space-efficient than random location storage, results in slightly higher throughput. [13] demonstrated that a velocity-based storage policy, where popular items are stored closer to pick stations, reduces travel distance by up to 10% compared to a random storage policy.

2.4.1 Pod Selection

Online retailers now have a solution to reduce labor costs and boost order picking efficiency due to the continuous development of the Robotic Mobile Fulfillment System (RMFS) in recent years [9]. In an RMFS, robots transport racks—or “pods,” as they are sometimes called to picking stations, where pickers take Stock Keeping Units (SKUs) out of the racks. The racks are then either moved to different picking stations or returned to the storage area by the robots. This automated system streamlines the order picking process, which is essential for fulfilling customer orders promptly [11].

The heart of RMFSs lies in their order picking process, as it plays a pivotal role in ensuring timely and accurate order completion. By strategically optimizing the sequence of orders and rack movements for each workstation companies can substantially boost the productivity of their picking operations [3]. However, the optimization of order picking in RMFSs presents a

challenging problem due to the complexity of coordinating robot movements and the dynamic nature of order demands.

This research aims to address the challenges associated with order picking in RMFSs by developing and evaluating optimization strategies that improve operational efficiency. By focusing on the sequencing of pods, the study aims to improve the performance of RMFS, thus enhancing warehouse management practices in the fast-growing e-commerce industry, making them more effective and cost-efficient.

2.4.2 Pod Retrieval

A crucial element of an efficient and successful inventory management system is the Warehouse Management System (WMS), of which the pod retrieval process is one. Pods are assigned to picking stations to facilitate the picking process. The pods are returned to storage once the pickers have finished their work. Retrieving pods efficiently is essential for maintaining inventory integrity and ensuring that customers receive their products on time.

There are multiple steps involved in the pod retrieval process: storage, picking, and pod assignment. During the pod assignment process, the Warehouse Management System (WMS) allocates a specific pod to a picking station based on inventory levels and the workload of the picker. The picking step involves the picker retrieving the allocated pod from the storage area and relocating it to the predetermined spot. The storage step involves returning the pod to its designated location in the storage area once the picker has completed its work.

In order to increase productivity and lower expenses, research has focused on streamlining the pod retrieval procedure. For instance, a study investigating the use of a rack retrieval system to enhance warehouse operations was published in the *Journal of Transportation Research Part E: Logistics and Transportation Review* [14]. The study discovered that by reducing the time needed to retrieve pods from storage, this system significantly improved retrieval efficiency.

In 2018, a separate study proposed a pod assignment algorithm that considers the workload of the picker and inventory levels to optimize assignments. This study was published on the arXiv preprint server [15]. The algorithm reduced the number of pods needed for retrieval, enhancing the efficiency of the pod assignment process.

Even with these developments, more studies are required to continue enhancing the efficacy and efficiency of the pod retrieval procedure. Further productivity gains and cost reductions in warehouse operations can result from improved approaches and technologies in this field.

2.5 Optimizations

Assignment problems are a fundamental class of optimization challenges encountered across

various fields, including logistics, resource allocation, and scheduling. These problems involve optimally matching elements of two sets, such as tasks and resources, to minimize overall costs or maximize efficiency. Over the years, researchers have employed diverse optimization techniques to address these problems, with significant developments in both approach and computational efficiency. Traditionally, mixed integer programming (MIP) models have been widely used to formulate and solve assignment problems [16]. MIP models offer flexibility in representing complex constraints and objectives, making them suitable for a wide range of real-world scenarios. While MIP models provide a robust framework for problem formulation, they can become computationally intensive for large-scale problems, often requiring sophisticated solvers and significant processing time. This limitation has led researchers to explore more efficient algorithms for specific classes of assignment problems.

The Hungarian method, also known as the Munkres algorithm, has emerged as a powerful alternative for solving assignment problems. The transition from MIP models to the Hungarian method represents a shift towards more specialized and efficient problem-solving approaches. While MIP models offer generality, the Hungarian method leverages the specific structure of assignment problems to achieve optimal solutions more rapidly. This efficiency becomes particularly crucial in dynamic environments or when dealing with large-scale systems where quick decision-making is essential.

2.6 Warehouse Operation Considerations

RMFS comprises three primary components, each of which significantly influences the warehouse design [17]. System comprises three primary components that significantly influence warehouse design. Firstly, the hardware components include autonomous mobile robots, which are central to the system. The warehouse layout must accommodate their mobility patterns, maintenance areas, and charging stations. Pods, the modular storage units, require a reconfigurable and adaptable storage space, with design considerations for accessibility, weight limits, and pod sizes. Strategically positioned human-operated picking and replenishment stations are also necessary to maximize productivity and minimize robot travel distances.

Secondly, the control system plays a vital role. It must ensure safe robot navigation with sufficient space and visibility, and charging stations should be strategically placed to minimize downtime. Clear routes and intersections are needed to enhance robot mobility and reduce traffic congestion, while variable-width aisles and speed zones help regulate traffic flow and avoid congestion.

Thirdly, the process management software is crucial for seamless operations. This includes

designated locations for order processing, thoughtful placement of sensors and Internet of Things devices for data capture, robust wired connections and Wi-Fi coverage for instantaneous data exchange, and a centralized control center for monitoring and managing the entire RMFS operation.

To accommodate these elements, the RMFS warehouse design typically features a grid-like arrangement for improved robot mobility, easily reconfigurable storage facilities that adapt to changing demand patterns, and scalable infrastructure to support fleet growth and increased storage capacity in the future. Careful planning is essential to integrate these components and create an effective RMFS environment. Factors such as ceiling height for potential multi-level pod storage, floor conditions for smooth robot mobility, and appropriate lighting for both machine and human vision systems must be considered. Additionally, buffer zones and staging areas should be included in the warehouse design to efficiently handle peak times and system outages. This comprehensive approach ensures optimal performance, scalability, and adaptability to evolving e-commerce demands.

2.7 Research Comparison

We present a table summarizing recent research focused on order picking. This table includes studies from the current year and highlights various approaches. Beyond analytical studies utilizing queuing or Mixed-Integer Programming (MIP) models, researchers have employed diverse methodologies to examine the performance of different operational decision rules and storage policies. Assuming it can solve quickly, a MIP model is suitable for studying operational performance objectives, particularly for estimating the makespan of batch orders. Table 2.1 reveals that most research in Robotic Mobile Fulfillment Systems (RMFS) concentrates on pod allocation (selection). However, fewer studies address robot scheduling and sequencing, and even fewer consider pod repositioning. Our research contributes to this field by focusing on pod selection in conjunction with robot assignment and taking into account pod repositioning.

Table 2.1 Recent Literature Overview

Paper	Pod			Robot			System	Methodology
	allocation	sequencing	reposition	allocation	sequencing	scheduling		
“Introducing split orders and optimizing operational policies in robotic mobile fulfillment systems” [18]	✓	✓		✓			RMFS	MIP, heuristic
“Joint optimization of order sequencing and rack scheduling in the robotic mobile fulfillment system” [8]	✓	✓		✓			RMFS	MIP, heuristic
“Rack retrieval and repositioning optimization problem in robotic mobile fulfillment systems” [14]	✓			✓			RMFS	matheuristic, SA
“Order picking optimization with rack-moving mobile robots and multiple workstations” [19]	✓	✓					RMFS	MIP, heuristic
“Order picking optimization in a robotic mobile fulfillment system” [20]					✓		RMFS	MINLP, heuristic
“Order and rack sequencing in a robotic mobile fulfillment system with multiple picking stations” [21]	✓	✓					RMFS	MILP, heuristic
“Optimization models for scheduling operations in robotic mobile fulfillment systems” [22]	✓				✓	✓	RMFS	MIP, simulation
“An efficient multi-agent approach to order picking and robot scheduling in a robotic mobile fulfillment system” [23]	✓			✓			RMFS	simulation
“A Three-Stage model for Clustering, Storage, and joint online order batching and picker routing Problems: Heuristic algorithms” [24]							RMFS	Three-stage mathematical model, heuristic
“How to deploy robotic mobile fulfillment systems” [17]	✓	✓	✓	✓			RMFS	MIP, heuristic
“A column generation driven heuristic for order-scheduling and rack-sequencing in robotic mobile fulfillment systems” [25]	✓	✓					RMFS	heuristic
“Online joint optimization of pick order assignment and pick pod selection in robotic mobile fulfillment systems” [26]	✓						RMFS	MIP, heuristic
“Stronger mixed-integer programming-formulations for order- and rack-sequencing in robotic mobile fulfillment systems” [27]	✓	✓					RMFS	MIP, heuristic
“Optimal resource allocation and routing in robotic mobile fulfillment systems” [28]	✓	✓		✓		✓	RMFS	MIP, heuristic (VNS)

CHAPTER 3

METHODOLOGY

3.1 Problem Definition and Research Framework

The purpose of this study is to illustrate the order-picking challenge in a warehouse using a RMFS. Order picking involves fulfilling every order placed by a consumer. Robots are utilized in the RMFS warehouse to transport goods to the storage area instead of human pickers. Pods need to be retrieved from the storage area and delivered to the replenishment or picking station by the robots. The robots return the pods to the storage area once they have completed their tasks at the station.

Order arrivals and pod arrivals are just two of the operations that constitute the order-picking process in the RMFS warehouse. Pods are selected for the picking station in this study, ensuring that the appropriate pods are chosen based on the order's specifications.

Every picking station in the warehouse has a maximum capacity. Each employee handles one pod at a time, considering the appropriate machine and path distance. Consequently, some scheduled pods may be left waiting in the storage area at any given processing time. Path considerations treat machines and pods as a single unit. Additionally, planned shelves are not factored into the decision-making process; only unscheduled shelves are considered when selecting a shelf. After an employee finishes examining the shelves, the system returns them to the pod retrieval area, changes their status to unscheduled, and includes them in the decision-making process again. When new orders are assigned to the station, the Warehouse Management System (WMS) updates this data in real-time and makes decisions accordingly.

Picking stations in the RMFS warehouse have a limited number of boxes, and each box can only be used for one order. Therefore, the total box capacity of a pickup station must not exceed the number of orders assigned to it. Each incoming order is scheduled based on the available case release time at each picking station, with each order being assigned to only one picking station. Orders arrive continuously, many of them containing numerous items that often exceed the capacity of a single pod. Additionally, each packing station has a restricted amount of space for employees to prepare orders. The pod-moving robots must collaborate effectively with the pod selection process.

It is thought that picking each object, regardless of size, requires the same amount of time to ensure efficiency. The recovery station was kept busy, and the researchers ensured that the appropriate boxes were available when needed. Multiple pods are used to distribute items, so selecting which pods to send to which stations to meet order criteria requires careful consideration. Choosing the correct pods remains the top priority, but decisions are made on packing one order at a time or handling several orders simultaneously, considering the assignment of robots to jobs.

Order completion time, overall system throughput, picking station utilization, and the number of pods utilized per order are used to evaluate the effectiveness of the suggested strategies. Two scenarios are presented to explore how mobile robots and pods can collaborate to optimize the warehouse operation. Definition of the Problem: The main challenge is figuring out how to route and choose the best pod configuration to process incoming orders in an RMFS. This problem is made more difficult by multiple factors.

This study will adhere to the Figure 3.1 research structure. The proposed model will consider robots that couple and pick shelves using similarity-based allocation methods. The study will be divided into two sections: robot route optimization and model optimization based on order scheduling. These two processes are interconnected within the system and can be carried out simultaneously. The system will utilize the Jaccard index during order placement to assess the similarity between orders. Shelves will be numbered based on the proximity of their SKUs to maximize shelf utilization. The number of shelves assigned at once will be determined by the complexity of the order. The system will then use the Manhattan distance to calculate the distance between the robot's location and the racks. Each assignment will be made based on the shortest path and the highest rack pick rate, ensuring that each decision is allocated to a single picking station.

The goal aims to develop a robust algorithm for pod selection in RMFS warehouses. The algorithm will adapt to dynamic order compositions and fluctuating order arrivals. Its goal is to balance picking station efficiency and minimize pod usage. And will also explore how different pod storage and replenishment strategies impact pod selection effectiveness. This will offer insights into how storage policies influence system performance and order fulfillment efficiency. The study will assess the scalability of these strategies across warehouses of varying sizes and order volumes. Additionally, the research will analyze how pod selection interacts with robot

routing strategies to optimize overall system performance. By studying these components together, the goal is to improve warehouse operations and order processing times.

Using a simulation-based methodology, researchers can test and refine pod selection algorithms under various scenarios. This approach allows for the exploration of complex system dynamics and the identification of optimal solutions without disrupting warehouse operations. By leveraging simulation techniques, innovative algorithms can be developed to enhance RMFS efficiency and effectiveness in real-world applications.

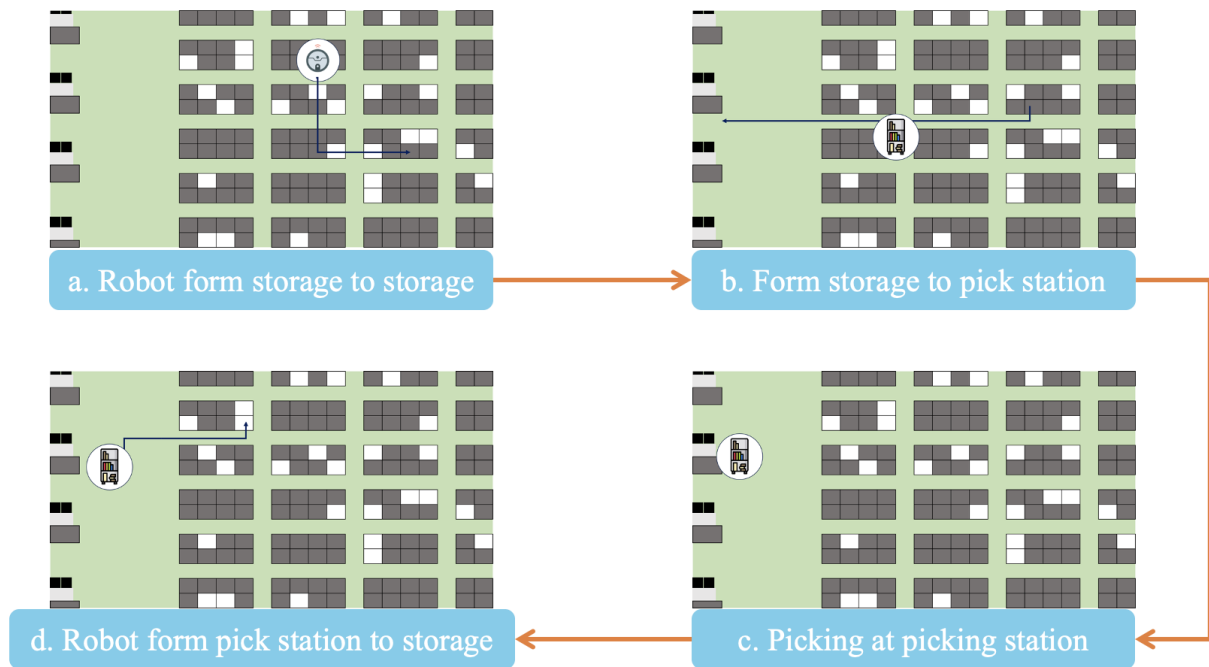


Figure 3.1 The Robot Pair's Traveling Cycle with The Pod [29]

3.2 System Process Flow

In Robotic Mobile Fulfillment Systems (RMFS), storage area organization is fundamental. Shelves can be arranged using fixed locations or random arrangements, with fixed locations generally incurring higher costs. An alternative approach, the Demand rule, selects pods based on current order backlog, prioritizing those with the most demanded units. Shelf slots are optimized to maximize capacity based on item dimensions, and inventory levels are calculated according to item frequency. A mixed integration policy and ABC classification (60% class A, 30% class B, 10% class C) are employed to optimize pod utility and ensure diverse SKU distribution within pods. This strategy aims to reduce replenishment frequency and is typically implemented before

system startup.

RMFS operations commence with an initial backlog of orders received during non-operational periods, with new orders continually arriving. The Warehouse Management System (WMS) guides pickers through displays, indicating which items to pick from specific shelf locations and where to place them for each order.

Each picking station has a fixed number of bins, and it's crucial to ensure empty bins are available before processing subsequent orders. Order allocation may be based on delivery deadlines, First-In-First-Out (FIFO) principles, batching, or other criteria. When an order reaches a workstation, the system generates a list of shelves capable of fulfilling it.

Recent research has explored various optimization strategies. Xie et al. (2021) [17] focused on order-workstation assignment and shelf selection to minimize the set of shelves used while considering order splitting among workstations. Zhuang et al. (2022) [10] addressed order and shelf sequencing for multiple workstations, aiming to minimize shelf movements while balancing workload and managing shelf conflicts. Some researchers, like Jiang et al. (2020) [30], have integrated order picking optimization with inventory storage assignment, recognizing that both factors influence order and shelf sequences. It's worth noting that while shelves can fulfill multiple requests, they are not considered for new tasks until they return to the storage area.

Once shelves are selected, robots transport them to target workstations. Robot deployment is based on their status (idle or returning to storage). Selection for tasks typically uses random assignment or shortest distance criteria. A hierarchy of movement priorities exists: robots carrying shelves to workstations have top priority, followed by those returning shelves. Robots without shelves can move more freely, including under shelves in any direction. Warehouse aisles are usually one-way, with specific rules for resolving conflicts at intersections.

Robots not only service picking stations but also replenishment stations, following the same routing rules. Replenishment is initiated after completing picking tasks. Shelves return to designated storage locations, with low-inventory shelves flagged and prioritized for replenishment. Autonomous Mobile Robots (AMRs) transport these shelves to replenishment stations based on various inventory decisions.

An important consideration is robot power management. Charging stations are strategically placed in the warehouse. When a robot's battery level falls below a safe threshold, it's scheduled for charging, which temporarily reduces the number of available robots in the system.

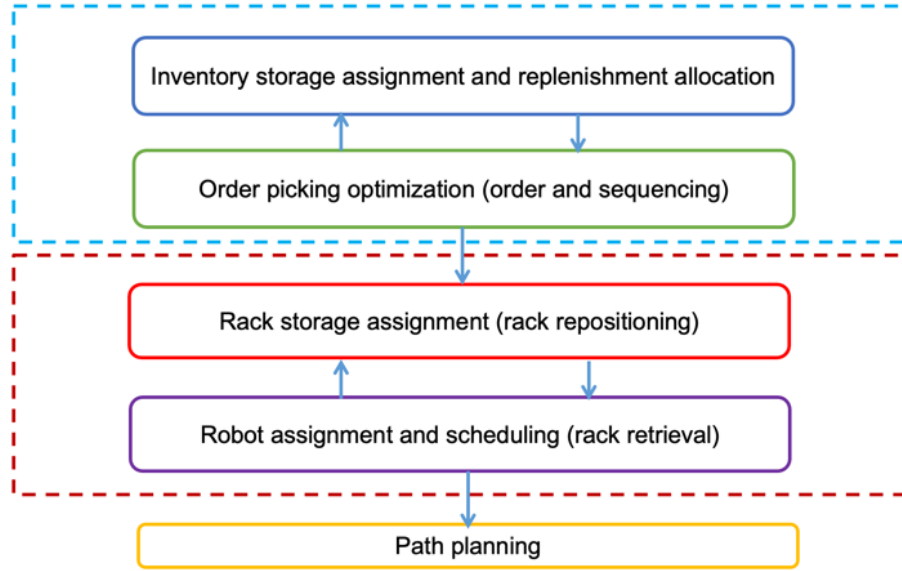


Figure 3.2 A Hierarchical Planning Process for RMFSs [14]

3.3 System Configuration

The Robotic Mobile Fulfillment System (RMFS) used in this study is designed to simulate a realistic warehouse environment. This section outlines the key components and parameters of the system configuration.

3.3.1 System Architecture

In this research, a comprehensive simulation and optimization environment for the Robotic Mobile Fulfillment System (RMFS) is developed using integrated platforms such as NetLogo, Python, and Microsoft Excel.

NetLogo is employed as the primary simulation tool to visualize the warehouse operating process. Agents can participate in dynamic interactions with one another in a virtual world, thanks to this robust modeling and simulation technology. These agents can be virtual entities such as supervisors overseeing the picking and restocking procedures, or they can represent a range of physical elements including workstations, storage pods, and Automated Mobile Robots (AMRs). NetLogo's interactive and visual features enable the comprehensive depiction and analysis of warehouse operations, providing users with a clear understanding of the system's dynamics.

Python is used as the programming language to manage the system functions and handle the complex logic required for optimization tasks. Processing the assignment results, making decisions

in real-time, and interacting with the simulation model in NetLogo are all handled by Python programs. This integration ensures that the system can effectively adapt to and respond to a variety of operational conditions by enabling seamless communication between the optimization algorithms and the simulation environment.

Microsoft Excel serves as the database management tool within this integrated platform. Data regarding layout design, orders, settings, pods, and goods are stored and managed using Excel. Excel simplifies data interchange and manipulation by importing inputs and exporting outputs from NetLogo and Python as CSV files. This setup allows the user to adjust and modify simulation parameters, analyze results, and conduct what-if scenarios by manipulating the data in Excel.

The system's design is illustrated in Figure 3.3, which also demonstrates how these three components collaborate. Typically, the process commences with setting up the operational parameters and warehouse layout in Excel. Subsequently, the simulation is executed by importing these configurations into NetLogo. Python scripts interact with NetLogo throughout the simulation to enhance task and resource allocation; real-time updates are made to the simulation based on optimization feedback. Once the simulation concludes, the operational insights and performance indicators are exported back to Excel for further reporting and analysis.

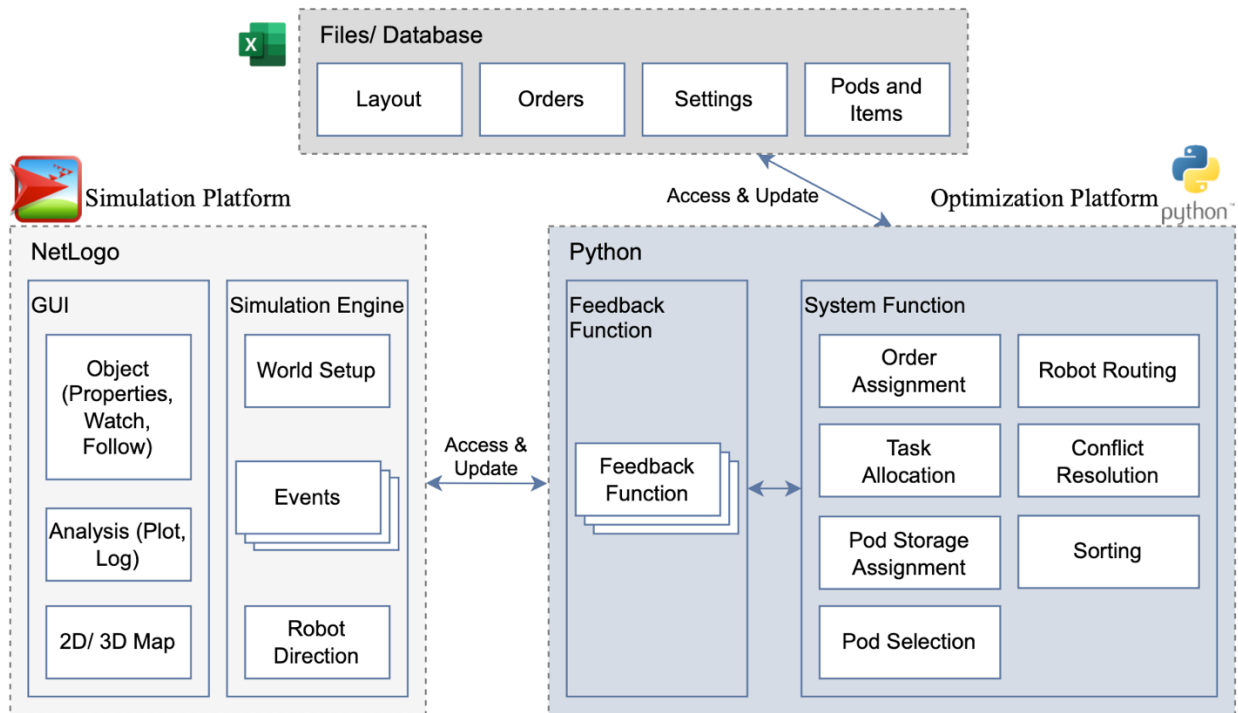


Figure 3.3 System Architecture

3.3.2 Simulation Layout

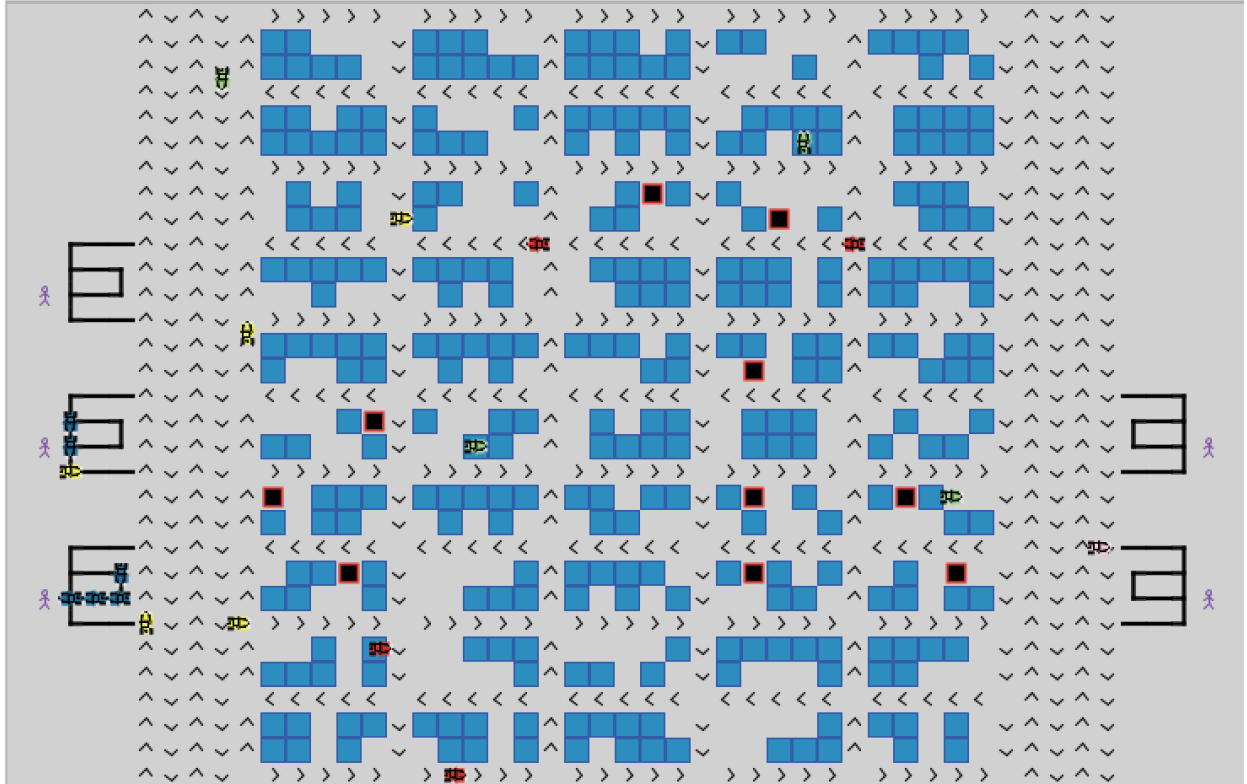


Figure 3.4 Simulation Layout

The research employs a simulation layout depicted in Figure 3.4, showcasing an integrated warehouse design within NetLogo. This layout encompasses picking stations, a storage area, and replenishment stations. On the left, picking stations with pickers and queue lines are positioned, while the central area houses pods containing SKUs. Robots, represented by various colors, indicate their current tasks: green for those assigned to pick target pods, red for robots carrying pods to picking stations, blue for those waiting in queue lines, yellow for robots returning to storage after picking, and pink for those heading to replenishment stations. Vacant storage spaces, shown as blank areas, offer alternative paths for robots to avoid conflicts. Charging stations are represented by black squares with red outlines. Autonomous Mobile Robots (AMRs) navigate one-way aisles and can move underneath pods without directional constraints. Within aisles, AMRs are limited to forward movement or stopping, but can turn at intersections. Near workstations, two-lane highways provide multiple routing options. This comprehensive layout design facilitates efficient robot and pod movement throughout the warehouse, optimizing order fulfillment processes while minimizing conflicts and delays.

3.3.3 Simulation Parameters

The simulation incorporates various factors to accurately replicate the full RMFS procedure, with Table 3.1 detailing the key parameters and assumptions. To minimize randomness, it runs for two hours with ten replications. Following industry standards, the ABC class rule is applied for SKU classification, and the warehouse layout mimics real-world RMFS practices by distributing items across multiple pods, ensuring no single pod contains all units of a particular item. Crucially, the study leverages generated data, enabling the modeling of hypothetical scenarios and the testing of new hypotheses unrestricted by real-world data limitations.

Table 3.1 System Parameters

Parameter	Value
Simulation	
Run length	2 hours
Replication	10 replications
Layout	
Inventory Area	500 locations
Inventory Capacity	300 pods
Empty Storage Area	190 locations
Pod Batch	2 x 5 blocks
Storage Area Aisles	6 vertical aisles
	11 horizontal aisles
	3 picking station
Station	2 replenishment station
	10 charging stations
Orders	
Initial Order	50 orders
Coming Rate	120 orders/per hour
Pods	
Capacity	10 slots
SKUs	
Number of SKUs	500 SKUs
SKU Distribution	Class A = 0-50 (10%)
	Class B = 51-200 (30%)
	Class C = 201-500 (60%)
SKUs/pod	10 SKUs (ABC Mixed Class)
Robot (AMR)	
Number of robots	20 robots
Robot Speed	1.5 m/s (maximum velocity)
	1 m/s ² (acceleration/deceleration)
Lift/Store the pod	2.2 s

Parameter	Value
Robot to pod policy	Shortest distance
Picking	
Bins	6
Queuing	9 robots
Time for handling	10 s
Picking Policy	Similarity-Based Order
Repelnishment	
Queuing	9 robots
Time for handling	20 s/ per SKU
Inventory Level	40%

3.4 Pod Selection Problem Formulation

Similarity-Based Order

$$J_{AB} = \frac{|A \cap B|}{|A \cup B|} \quad (3-1)$$

Notations

A : Set of SKUs contain in order A

B : Set of SKUs contain in order B

The Jaccard similarity coefficient provides a measure of how similar two orders are, which helps in grouping orders that can be efficiently fulfilled by the same pods.

Pod Selection Objective

Hungarian Method

$$\min C = \sum_{p \in P} \sum_{s \in S} \left(D_{ps} \cdot Sim_{ps} \cdot \frac{1}{SKU_{ps}} \right) \cdot X_{ps} \quad (3-2)$$

s.t

Total Distance Calculation

$$D_{ps} = d_{1,ps} + d_{2,rp} + d_{3,ps} \quad (3-3)$$

Similarity Score

$$Sim_{ps} = |R_p \cap O_s| \quad (3-4)$$

SKU Fulfillment Ratio

$$SKU_{ps} = \frac{\theta_p}{|O_s|} \quad (3-5)$$

Notations

C	: The cost of the Hungarian method
D_{ps}	: The total distance for pod p to serve station s
Sim_{ps}	: Similarity score between pod p and order at station s
SKU_{ps}	: SKU fulfillment ratio for pod p and at station s
X_{ps}	: $\begin{cases} 1 & \text{if pod } p \text{ is assigned to station } s, \\ 0 & \text{otherwise} \end{cases}$
$d_{1,ps}$	: Manhattan distance between pod p and picking station s
$d_{2,rp}$	: Manhattan distance between robot r and storage location p
$d_{3,ps}$	: Manhattan distance from the picking station s back to storage location p
R_p	: Set of SKUs contained in pod p
O_s	: Set of SKU and quantity of a new incoming order at the station
θ_p	: The number of the SKU type fulfilled in pod

The Hungarian method's advantages are capitalized upon by the pod selection strategy, which also takes into account the intricate, many-to-many structure of the warehouse order fulfillment process. In order to ensure effective order processing and resource usage, this technique aims to assign pods to picking stations in the most efficient manner possible.

Regarding the pile-on impact that was previously discussed, it is important to note that our method for making decisions already includes a mechanism for giving stations that have a larger potential for pile-on a higher similarity. This is reflected in the similarity index, which takes into account the intersection of the SKUs in the order and those in the pod. An SKU fulfillment ratio has also been established to further prioritize the assignment of pods that contain the appropriate amounts of key products. This guarantees that pods with sufficient stock levels will be given priority during the allocation process.

$$\sum_{p \in P} X_{ps} \leq 1, \forall s \in S \quad (3-6)$$

$$\sum_{s \in S} X_{ps} \leq 1, \forall p \in P \quad (3-7)$$

$$\sum_{o \in OP_i} |o| \leq CA_i \quad (3-8)$$

$$TP_i \leq T, TP_i \geq 0 \quad (3-9)$$

$$\sum_{k \in I_K} q_{r,K} \geq Q_{o,K}, \forall k \in K \quad (3-10)$$

$$\sum_{r \in R} \sum_{p \in P} Y_{rp} = 1 \quad (3-11)$$

Notations

i	: Number of Picking Station
CA	: The capacity of the bins at a station
OP	: Set of orders assigned to each station
OP_i	: Set of orders assigned to picking station i
T	: Number of bins in each station
TP_i	: Number of bins available in the station
I_K	: Set of the pod containing SKU K
$q_{r,K}$	: Quantity of SKU K in pod r
$Q_{o,K}$	: Required quantity of SKU K in order o
K	: Set of all SKUs
R	: Set of available robots
Y_{rp}	: $\begin{cases} 1 & \text{if robot } r \text{ is assigned to pod } p, \\ 0 & \text{otherwise} \end{cases}$

Additional

n	: Number of orders
m	: Number of pods

Equations 3-6 to 3-7 prevent double assignments, ensuring that each station pod is assigned to only one station, and that each bin is allocated to a single location within the station's limited bin slots. Equation 3-9 and 3-10 addresses bin availability, limiting the number of bins that can be assigned to each station. SKU quantity requirements are captured in equation 3-11, guaranteeing that the required quantity of each SKU is met. Each robot is assigned to a pod retrieval task and will stay with the pod until the picking operation is completed and the pod is stored back to the storage area.

3.4.1 Pod Selection Model

For a comprehensive understanding of the order handling process, please refer to Figure 3.2, which illustrates the system flowchart in detail. This flowchart provides a visual representation of the sequential steps and decision points involved in processing an order through our system, from the initial receipt to final fulfillment.

The Pod Selection Model in our RMFS employs a sophisticated multi-stage process to optimize order fulfillment and pod utilization. Initially, the system evaluates the total bin capacity and availability across all stations, mapping current order assignments to available resources. A critical decision point occurs during order assignment to stations. The system prioritizes orders based on available bin capacity, implementing a hold mechanism for excess orders until the next assignment period. This ensures efficient use of station resources. The assignment process incorporates a similarity index, a key decision factor that determines the most suitable station for each order. When a clear match isn't found, the system makes a strategic decision to randomly allocate the order, maintaining operational flow.

The pod selection process introduces several important decision points. First, the system assesses pods already en route to stations, updating the remaining order set. This real-time adjustment is crucial for maintaining an accurate inventory picture. The system then generates a new set of incoming orders and makes the important decision of selecting assigned pods with matching SKU types, initiating the core of the pod selection algorithm.

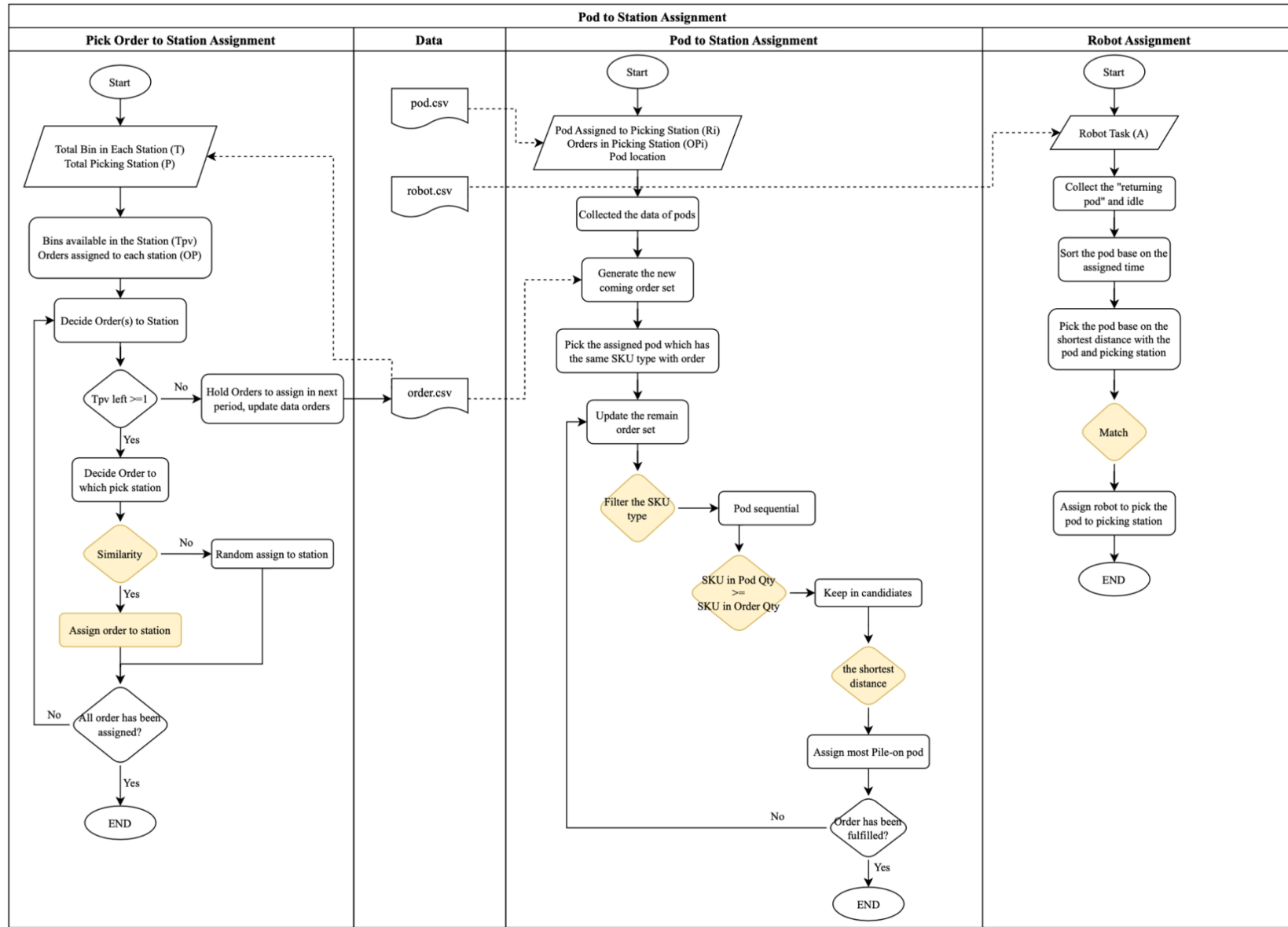
A pivotal decision in the model is the implementation of the Hungarian method to solve the many-to-many assignment problem between candidate pods and orders. This choice is significant as it provides a simple yet powerful solution to a complex allocation challenge. The cost function used in this method is carefully designed to include not just distance or time metrics, but also a similarity score. This score reflects how well a pod's contents match the order requirements, promoting efficient pod usage by encouraging the selection of pods that can fulfill multiple items from a single order. The inclusion of the similarity score in the cost function represents a strategic decision to enhance pod usage efficiency. By prioritizing pods that can fulfill more items per order, the system aims to reduce the total number of pod movements and increase the items picked per pod visit to a station. This decision directly addresses the goal of optimizing pod utilization, a critical factor in overall system efficiency.

Furthermore, the model incorporates a "fulfillment" metric in its decision-making process.

This metric evaluates whether a pod's quantity can fully meet an order's needs, adding another layer to the pod selection criteria. Pods meeting these criteria are added to a candidate list, forming the basis for the Hungarian method's optimization. The final stage involves coordinating pod selection with robot assignments. The system decides which available robots (either returning or idle) to pair with selected pods, prioritizing shortest travel distances. This integrated decision-making process ensures that pod selection, order fulfillment, and robot movement are optimized collectively.

By incorporating these decision points, particularly the use of the Hungarian method with a similarity-enhanced cost function, the model achieves a balance between maximizing picking station efficiency and optimizing pod usage. This approach allows for dynamic adaptation to varying order compositions and warehouse conditions, ultimately leading to more efficient RMFS operations.

Figure 3.5 System Flowchart for Model



3.4.2 Compare Model

The compare model in this study is based on the Random pod selection rule, which serves as our baseline. This choice is informed by the findings of Merschformann (2019) [31], who evaluated various pick pod selection (PPS) strategies. While Merschformann's results showed Lateness as the best-performing rule in terms of throughput rate, followed by Random, Demand, Pile-on, Age, and Nearest, our system's constraints necessitate a different approach.

In our research context, we assume all orders will be fulfilled within one day, eliminating the need for strategies that prioritize order urgency or lateness. The Random rule in our baseline model randomly selects a pod from the storage area, filtered to contain at least one SKU type matching the order requirements. This approach is straightforward and serves as a neutral starting point for comparison. It's important to note that due to the limited capacity of slots and the fixed dimensions of items, a single pod may not always contain sufficient quantity to fulfill an entire order. In such cases, multiple pods may need to be selected to complete the order.

Once a pod is selected, it is paired with a robot. The robot then uses the Dijkstra algorithm to determine the shortest path for picking up the pod and delivering it to the station. This path-finding approach optimizes the travel time for both the robot-to-pod and pod-to-station movements. For order processing, the compare model utilizes the same decision-making approach as the proposed model. This consistency in order handling allows for a fair comparison between the Random pod selection baseline and the more sophisticated strategies employed in the proposed model.

3.5 Performance Analysis

This study compares the effects of several simulation-generated scenarios using a number of indicators to assess the effectiveness of the warehouse simulation. The following notes relate to the performance measuring criterion.

$$Po = \frac{S_f}{|P|} \quad (3-12)$$

$$TP_{EF} = \frac{O_f}{O_t} \quad (3-13)$$

Notations

Po : Pile-on

S_f : The number of SKUs being picked

P	: The set of the pods visits
O_f	: The fulfilled set of orders
TP_{EF}	: Throughput efficiency
O_t	: Order rate in period t

To evaluate the effectiveness of our proposed pod selection algorithm, we employ three key performance metrics: pile-on value, system throughput, and throughput efficiency. These metrics provide a comprehensive view of the system's operational efficiency and its ability to handle order volumes effectively.

1. Pile-on Value (Equation 3-12):

The pile-on value measures the average number of SKUs picked from a single pod during the simulation runtime. This metric is crucial as it reflects the efficiency of our pod selection algorithm in consolidating picks and minimizing unnecessary pod movements. A higher pile-on value indicates better utilization of each pod visit to a picking station.

2. Throughput Efficiency (Equation 3-13):

Throughput efficiency is a more nuanced metric that considers the system's ability to handle incoming orders in real-time. It compares the system's throughput to the order arrival rate per hour. This metric provides a straightforward reflection of whether the system can keep pace with incoming orders, with a value close to or exceeding 1 indicating that the system is effectively managing or outpacing the order influx.

3.6 Sensitivity Analysis

To comprehensively evaluate the robustness and performance of our proposed pod selection algorithm, we employ the Taguchi method for sensitivity analysis. This approach is particularly well-suited for our complex RMFS simulation model, which involves multiple interacting factors that can influence system performance. Sensitivity analysis offers several advantages for our study: it allows us to assess how well our algorithm maintains performance under various operational conditions and parameter changes, helps pinpoint the most influential factors affecting our algorithm's performance, and guides future optimization efforts. Additionally, it provides a structured way to design experiments and analyze results, ensuring comprehensive coverage of the factor space. By systematically varying input parameters, we can evaluate the simulation's robustness to changes in these parameters, validate the accuracy of our simulation results, and test

the model's assumptions and input parameters. We focus on two primary response variables: the pile-on value and throughput efficiency, providing a balanced view of both the micro-level efficiency (pod utilization) and macro-level performance (overall system throughput) of our algorithm. Conducting sensitivity analysis by modifying several key parameters of the RMFS model allows us to determine how much these parameters can be changed without significantly affecting the objective values, providing insights into the model's stability and the algorithm's adaptability. This information is crucial for both further development of the algorithm and its practical implementation in real-world RMFS environments.

CHAPTER 4

RESULTS AND DISCUSSION

The pod to station assignment discussed in this study is a fundamental problem of RMFS. Therefore, any warehouse implementing RMFS must face this problem. In this study, the input simulation is based on real case data. This chapter presents the results of our simulation model, focusing on the performance of the baseline system and our proposed pod selection strategy. We begin by examining the key performance metrics of the baseline model, followed by a detailed analysis of our pod selection algorithm's impact on system throughput and efficiency. The chapter concludes with a comparative discussion of these two approaches, highlighting the improvements achieved through our strategy. The following sections provide a detailed examination of the data used in our research, the results obtained from our simulations, and a comprehensive analysis of these findings.

4.1 Data

This section provides a detailed explanation of the data used in this research. To run the simulation effectively, several datasets are required: database orders, orders, and order finished dataset. These datasets form the foundation of our analysis and are crucial for understanding the performance of both the baseline model and our proposed pod selection strategy.

4.1.1 Data Variables

The dataset used in this research is a generated warehouse dataset. This approach allows us to control and manipulate various parameters to test our models under different conditions. The data are generated following specific parameters designed to mimic real-world warehouse operations. The database orders, which form the core of our simulation, consist of several key variables. These variables are listed and described in Table 4.1.

The order data generation process employs two stochastic methods to simulate realistic order patterns. First, the order arrival rate is modeled using a Poisson distribution. This statistical method is particularly suitable for simulating the number of orders arriving per unit of time, as it effectively captures the random nature of order arrivals in a warehouse setting. The Poisson distribution is characterized by a single parameter, λ (lambda), which represents the average number of orders per unit time. Second, the number of items per order is modeled using a Geometric distribution.

This distribution is appropriate for modeling the variability in order sizes, as it represents the number of trials needed to achieve a first success in a series of independent trials. In the context of order generation, it simulates the probability of an order containing a certain number of items, allowing for a realistic representation of order size variability.

Table 4.1 Order Database Characteristics

Category	Column Name	Description	Data Value
Database order	order_id	The sequence of the order, which base on the coming time	$\mathbb{Z}$
	order_dum	The sequence of the order, which base on the coming time	$\mathbb{Z}$
	order_type	Differentiate the order type, picking & replenishment	[1] picking [2] replenishment
	item	SKU ID	$[0, 1, 2, \dots, K]$
	qty	Quantity of SKUs that are being ordered	$[-1]$, no order assigned
	facing	The facing of the pod where the order being assigned	$[-1]$, no order assigned $[0,1]$, order assigned
	due_date	Order should be finished before the due date	
	station	The station that chooses based on the order	$[1, 2, \dots, i]$
	pod_id	Pod status related to the order	$[-1]$ assigned pod
	status	Order Status	$[-3]$, not assigned
	finish_time	The time when the order is already being fulfilled	$[-1]$, not assigned
	date	The date of the order	
	time_gen	The time when the order was coming to simulation to processed	$\mathbb{R}$

These variables allow us to simulate a variety of order scenarios, enabling a comprehensive analysis of our pod selection strategy under different conditions. In addition to the basic order information, our dataset includes detailed order allocation characteristics. Table 4.2 presents these characteristics, which are crucial for the picking process simulation.

Table 4.2 Assigned Order Characteristics

Category	Column Name	Description	Data Value
Assigned order	sequence_id	The sequence of the SKU after the order created in the simulation	$\mathbb{R}$
	order_id	The sequence of the order, which base on the coming time	$\mathbb{Z}$
	order_type	Differentiate the order type, picking & replenishment	[1] picking [2] replenishment
	item_id	SKU ID	$[0, 1, 2, \dots, S]$
	item_quantity	Quantity of SKUs that are being ordered	$\mathbb{R}$
	order_arrival	The time when the order is coming	$\mathbb{Z}$
	assigned_station	The order processed in which station	$[1, 2, \dots, i]$
	assigned_pod	The pod number to get the assignment by the order	$[0, 1, 2, \dots, m]$
	status	Pod status related to the order	“[-3] hasn’t assign pod [-1] assigned pod [0] pod queuing in station [1] finished the picking”
	order_processed	The order assigned to station time	$\mathbb{R}$
	order_finished	The time when the SKU was fulfilled	$\mathbb{R}$

The assigned order data provides comprehensive information about the picking process. It specifies which order is assigned to each picking station, identifies the numbered pod where the items for the order are picked, and records critical time points such as when the order is assigned to a workstation and when it is scheduled for fulfillment. This detailed information enables us to accurately simulate and analyze the entire order picking process, from assignment to completion.

4.2 Simulation Model Results

This section presents the results of our simulation model, comparing the performance of the baseline model with our proposed pod selection strategy. We conducted 10 replications for each

model to ensure reliability and statistical significance in our findings.

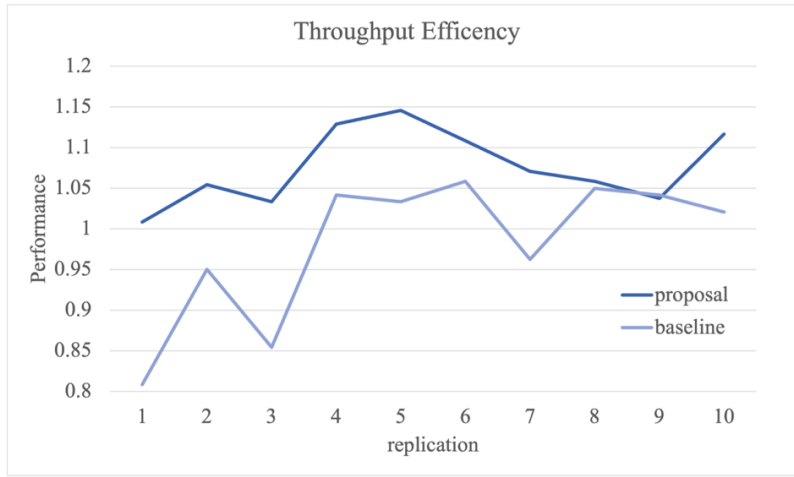


Figure 4.1 Performance Comparison of Baseline and Proposed Models (TP_{EF})

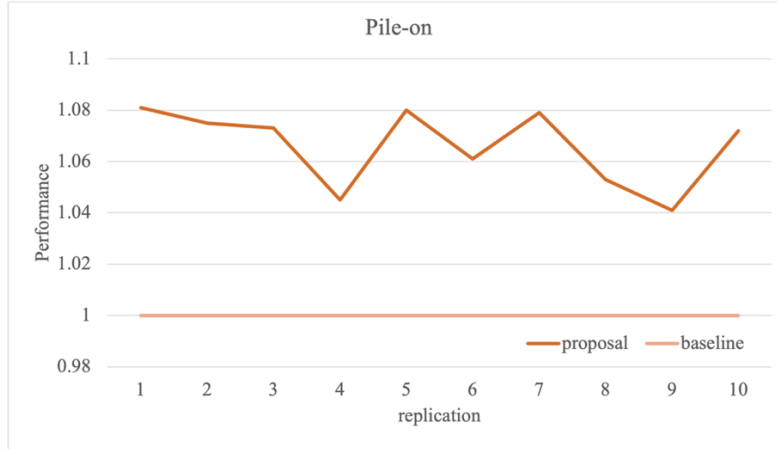


Figure 4.2 Performance Comparison of Baseline and Proposed Models (Po)

Table 4.3 Performance Metrics

TP_{EF}	Proposal	Baseline
Mean	1.07625	0.98208
Variance	0.00212	0.00774

As evident from Table 4.3 and Figure 4.1, our proposed model demonstrates superior performance compared to the baseline model. The proposed model achieves a higher mean performance compared to the baseline model, indicating better overall efficiency. To rigorously assess the significance of the observed performance difference, we conducted a Paired Two Sample t-test for Means. We chose a one-tailed test as our hypothesis was that the proposed model

would outperform the baseline model. The results of this statistical analysis are as Table 4.4.

Table 4.4 Paired T-test Analysis of Performance Metrics

	Proposal	Baseline
Mean	1.07625	0.98208333
Variance	0.00212211	0.00773553
Observations	10	10
Pearson Correlation	0.69411919	
Hypothesized Mean Difference	0	
df	9	
t Stat	4.57690393	
P(T<=t) one-tail	0.00066685	
t Critical one-tail	1.83311293	

The t Stat value (4.577) being greater than the t Critical one-tail value (1.833), combined with a very low p-value (0.0006), provides strong evidence that our proposed model significantly outperforms the baseline model. This means there is a very low probability that the observed improvement occurred by chance. This analysis supports our hypothesis that the proposed pod selection strategy offers a statistically significant improvement over the baseline model, based on the 9 paired simulation replications. By using the one-tailed test results, we're directly addressing our hypothesis that the proposed model outperforms the baseline, which aligns better with the goals of your study. This approach provides a more focused and powerful statistical argument for the effectiveness of your proposed strategy.

4.3 Sensitivity Analysis

The study perform a sensitivity analysis to identify the key variables that may influence the performance of our pod selection algorithm within the RMFS model. The primary variables considered are the number of robots, picking station availability, and order arrival rates. To comprehensively understand the impact of each factor on system performance, we define three levels for each variable. The first level corresponds to the parameters used in our proposed model. The other levels are determined by making minor adjustments to the hardware setup, which are cost-effective yet capable of revealing significant performance variations.

4.3.1 Factor Identification and Level Selection

To achieve a systematic and efficient exploration of these factors, we employ the Taguchi

method, utilizing an orthogonal array (OA) to design our experiments. Specifically, we select the L9 orthogonal array, which is well-suited for testing three factors at three levels each. The use of an OA ensures that we achieve a balanced representation of all possible combinations of factor levels, allowing us to conduct a comprehensive analysis with a minimal number of experimental runs. By employing the Taguchi method, we can effectively study the interaction between different factors and identify the most influential variables affecting the algorithm's performance. This approach reduces the need for exhaustive simulations while still providing robust and reliable results.

We systematically collect data on two primary response variables: the pile-on value and throughput efficiency. The detailed settings for the levels of each factor are presented in Table 4.5, while the results of the experiments are summarized in Tables 4.7 and 4.8. This structured approach ensures that we can accurately identify and quantify the effects of key variables on the performance of our pod selection algorithm.

Table 4.5 Levesl of Each Factor

Factor	orders	stations	robots
Level 1	120	3	20
Level 2	120	2	18
Level 3	160	4	22

Table 4.6 Design Orthogonal Array

	orders	stations	robots
1	120	3	20
2	120	4	22
3	120	2	18
4	140	4	20
5	140	2	22
6	140	3	18
7	160	2	20
8	160	3	22
9	160	4	18

Table 4.7 Empirical Data of Pile-on

	run1	run2	run3	run4	run5	run6	run7	$\bar{y}$	S
1	1.081	1.075	1.073	1.08	1.061	1.079	1.053	1.07	0.010
2	1.08	1.078	1.068	1.081	1.058	1.068	1.063	1.07	0.008
3	1.064	1.067	1.077	1.071	1.048	1.102	1.057	1.07	0.016
4	1.066	1.061	1.086	1.093	1.069	1.051	1.066	1.07	0.013
5	1.054	1.061	1.107	1.064	1.042	1.049	1.049	1.06	0.020
6	1.048	1.066	1.12	1.085	1.056	1.053	1.061	1.07	0.023
7	1.06	1.076	1.073	1.042	1.049	1.074	1.039	1.06	0.015
8	1.055	1.075	1.076	1.044	1.052	1.064	1.03	1.06	0.015
9	1.077	1.058	1.079	1.053	1.056	1.081	1.054	1.07	0.012

Table 4.8 Empirical Data of Throughput Efficiency

	run1	run2	run3	run4	run5	run6	run7	$\bar{y}$	S
1	0.975	1.008	1.054	1.146	1.108	1.071	1.058	1.06	0.053
2	1.154	1.167	1.050	1.171	1.138	1.075	1.150	1.13	0.044
3	0.850	0.858	0.842	0.954	0.975	0.875	0.983	0.91	0.058
4	0.989	0.989	0.936	1.014	1.107	0.986	0.914	0.99	0.057
5	0.804	0.854	0.796	0.825	0.864	0.857	0.779	0.83	0.031
6	0.850	0.861	0.796	0.839	0.896	0.857	0.779	0.84	0.037
7	0.644	0.672	0.734	0.700	0.700	0.706	0.756	0.70	0.034
8	0.838	0.703	0.847	0.825	0.825	0.816	0.800	0.81	0.045
9	0.756	0.700	0.863	0.816	0.803	0.878	0.825	0.81	0.057

4.3.2 Analysis of Means and Variances

Following the data collection, we analyze the means and variances of the response variables to determine the main effects of each factor. This analysis helps in identifying which factors have the most significant impact on the performance of our pod selection algorithm. By comparing the means of different levels of each factor, we can pinpoint optimal levels that enhance system performance.

Y1	Level1	Level2	Level3	DOF	SS	V	F	confidence	rho
orders	1.07	1.07	1.06	2	0.00	0.00	2.08	86.52%	6.85%
station	1.07	1.06	1.07	2	0.00	0.00	0.63	46.32%	2.07%
robots	1.07	1.07	1.06	2	0.00	0.00	0.63	46.10%	2.06%
	1.07	1.07	1.07	2	0.00	0.00	0.02	2.44%	0.08%
erro				54	0.01	0.00	1		88.93%

Figure 4.3 ANOVA of Pile-on

Fig 4.3 presents the results of the analysis of variances (ANOVA) for three factors: orders, station, and robots, each evaluated at three levels. The primary response variable analyzed is the pile-on value (Y1). The table includes degrees of freedom (DOF), sum of squares (SS), variance (V), F-value (F), confidence level, and percent contribution (rho%). The ANOVA results indicate that the factor orders has the highest F-value (2.08) and the highest confidence level (86.52%), suggesting that it is the most influential factor affecting the pile-on. However, the F-value is still below the typical critical value for statistical significance, indicating that while orders arrival rate might have the largest relative impact among the factors tested, the effect is not strong enough to be statistically significant on its own. The station availability and number of robots factors have lower F-values (both 0.63) and lower confidence levels (46.32% and 46.10%, respectively), indicating that their impact on the throughput efficiency is less pronounced and not statistically significant within the tested ranges.

The most striking observation is the very high error percentage (88.93%). This suggests that most of the variation in the experiment is not explained by the factors we're controlling (orders, stations, robots). Orders have the highest contribution at 6.85%, though this is still quite low. Stations and Robots have very similar, low contributions (2.07% and 2.06% respectively). Given that all factors have relatively low contributions compared to the error term, pooling might not provide much benefit in this case.

Y1	Level1	Level2	Level3	DOF	SS	V	rho
order	1.03	0.89	0.77	2	0.71	0.36	57.51%
stations	0.90	0.81	0.98	2	0.30	0.15	24.25%
robots	0.92	0.85	0.92	2	0.07	0.04	5.72%
	0.90	0.91	0.89	2	0.00	0.00	0.20%
erro				54	0.15	0.00	12.33%

Figure 4.4 ANOVA of Throughput Efficiency

In analyzing the factors affecting the performance of our pod selection algorithm in the RMFS model, we identified the following key influences and their contributions: order is the most significant factor, contributing 57.51% (rho) to the variation in results. Stations follows as the second most influential factor, contributing 24.25%, and Robots is the least influential among the controlled factors, with a contribution of 5.72%. Order has an SS of 0.71 and a V of 0.36, indicating substantial variability attributed to this factor. Stations has an SS of 0.30 and a V of 0.15, showing a moderate effect. Robots has an SS of 0.07 and a V of 0.04, reflecting its relatively minor influence.

Each factor has 2 degrees of freedom (DOF), suggesting a 3-level design for each factor, likely using an L9 orthogonal array. This design framework helps systematically examine the effects of each factor with minimal experimental runs. The error term in the analysis has a contribution (rho) of 12.33% and 54 DOF, indicating that the experiment included multiple replications or a larger experimental setup. This low error contribution suggests that the experiment has successfully captured most significant sources of variation.

Examining the level effects, we find:

- Order: Level1 (1.03) > Level2 (0.89) > Level3 (0.77)
- Stations: Level3 (0.98) > Level1 (0.90) > Level2 (0.81)
- Robots: Level1 (0.92) = Level3 (0.92) > Level2 (0.85)

4.3.3 Optimal Parameter Combination

Robots and the previously unidentified factor have been pooled into the error term. As shown in Fig 4.5, Order and Stations remain the two most influential factors. Now, the degrees of freedom (DOF) have increased to 58, reflecting the pooling of insignificant factors. The high F-ratios and 100% confidence levels for both Order and Stations indicate that these factors are highly significant and their effects are not due to chance. This strengthens the reliability of the findings. Pooling the less significant factors has helped to clarify the analysis by emphasizing the truly important factors. The increase in the error term's contribution (from 12.33% to 18.24%) is acceptable given the clearer picture it provides. For Order, Level 1 (1.03) appears optimal. For Stations, Level 3 (0.98) seems optimal. Investigate the characteristics of this station setup that make it superior. Thus, we will conduct confirmation runs using the optimal settings (Order: 120, Stations: 4, Robots: 20) to verify the predicted improvements.

Y1	Level1	Level2	Level3	DOF	SS	V	F	confidence	rho
order	1.03	0.89	0.77	2	0.71	0.36	91.41	100.00%	57.51%
stations	0.90	0.81	0.98	2	0.30	0.15	38.55	100.00%	24.25%
robots									
erro				58	0.23	0.00	1		18.24%

Figure 4.5 Pooled Table of Throughput Efficiency

4.4 Discussion

The analysis of our proposed model yields several important insights into the optimization of complex systems. Our statistical analysis demonstrates that the proposed model outperforms the baseline, indicating a significant improvement in system performance. The application of the Taguchi method to analyze three key factors - Orders, Stations, and Robots - at three levels each, revealed nuanced influences on system performance metrics. A critical finding emerged regarding the pile-on value, a key measure of system efficiency. The analysis showed that among the factors studied, the number of orders exerts the most significant influence, accounting for 57.51% of the relative contribution. This result underscores the importance of order management in optimizing system performance. It suggests that strategies focused on order processing and management could yield substantial improvements in overall system efficiency.

Interestingly, our analysis of throughput efficiency painted a different picture. None of the three main factors showed a significant influence on this metric, with the error term accounting for a striking 88.93% of the variation. This unexpected result suggests the presence of other influential factors not captured in our current model. It highlights the complexity of the system and indicates that throughput efficiency might be governed by more subtle or complex interactions than initially hypothesized. Based on our analysis, we identified an optimal parameter combination of 120 Orders, 4 Stations, and 20 Robots. Confirmation runs using these settings yielded mixed results. The average pile-on value of 1.0700 represents a slight improvement, indicating a more controlled and efficient process. More notably, we observed an average throughput efficiency of 1.1140, representing an 11.4% increase over the baseline. This significant improvement in throughput efficiency is a strong indication of the potential benefits of our optimization efforts.

However, the discrepancy between predicted improvements and actual results in the confirmation runs warrants careful consideration. It underscores the challenges inherent in

optimizing complex systems and suggests that our model, while improved, may still not capture all relevant factors or interactions. This limitation presents an opportunity for future research, particularly in identifying and incorporating additional factors that influence throughput efficiency. First, there's a clear need to investigate other factors that might influence throughput efficiency, given the high unexplained variance in our current model. This could involve expanding the range of factors considered or exploring more complex interactions between existing factors. Additionally, refining our model to better predict outcomes could enhance its utility in real-world applications.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This research focused on optimizing pod selection within Robotic Mobile Fulfillment Systems (RMFS) to enhance order fulfillment efficiency in e-commerce warehouses. Through the development and implementation of a novel two-step pod selection algorithm, integrated with the Hungarian algorithm for robot-pod assignment, we aimed to address the challenges posed by increasing order volumes and complexity in automated warehousing environments.

The study employed extensive simulations to evaluate the proposed algorithm across various operational scenarios. The results demonstrate a statistically significant improvement in system performance compared to the baseline model. Specifically, the proposed model achieved a mean performance of 1.07625 compared to the baseline's 0.98208, representing a 9.59% improvement in overall efficiency. Statistical analysis using a Paired Two Sample t-test yielded a t-statistic of 4.577 (greater than the critical value of 1.833) and a p-value of 0.0006, providing strong evidence for the superiority of our proposed model. Sensitivity analysis using the Taguchi method revealed that the number of orders is the most influential factor on system performance, accounting for 57.51% of the relative contribution to the pile-on value, a key efficiency metric. The optimal parameter combination identified (120 Orders, 4 Stations, 20 Robots) resulted in an average pile-on value of 1.0700 and an average throughput efficiency of 1.1140, representing an 11.4% increase over the baseline in throughput efficiency.

These results underscore the potential of our proposed algorithm to significantly enhance RMFS performance in e-commerce warehouses. The substantial improvement in throughput efficiency, in particular, demonstrates the practical value of our optimization approach in meeting the increasing demands of modern e-commerce operations. However, our analysis also revealed areas for further investigation. The high unexplained variance (88.93%) in throughput efficiency suggests the presence of additional influential factors not captured in our current model. This finding highlights the complexity of RMFS optimization and presents opportunities for future research.

In conclusion, this study contributes to the growing body of knowledge in warehouse automation by providing a novel approach to pod selection optimization in RMFS. The significant

improvements observed in both pile-on value and throughput efficiency offer valuable insights for e-commerce businesses seeking to enhance their operational efficiency. The methodologies and findings presented in this thesis represent a meaningful step forward in RMFS optimization, offering both theoretical contributions to the field of warehouse management and practical applications for the e-commerce industry. As the demand for efficient order fulfillment continues to grow, the insights gained from this study will play a crucial role in shaping the future of automated warehousing systems.

5.2 Future Work

This study, while contributing significantly to the field, presents several opportunities for improvement and extension in future research. The current model's approach of selecting orders into SKUs based on intra-order SKU similarity could be enhanced by exploring order batching techniques and considering SKU affinity within pods. This refinement could potentially lead to more efficient picking strategies. To bolster the robustness and practical applicability of our findings, future studies should consider extending simulation durations and validating the model using real-world warehouse data. This would provide more realistic performance metrics and enhance the model's relevance to actual warehouse operations. While our study primarily focused on maximizing pile-on efficiency, future research would benefit from incorporating additional evaluation criteria, such as the number of tasks or makespan. This expansion would offer a more comprehensive view of system performance and address the crucial aspect of employee workload balance, a key consideration for companies. Further advancements could be made by employing mathematical modeling to solve the optimization problem or by utilizing metaheuristic approaches, particularly for optimizing the pairing of pods and robots to minimize task numbers. Additionally, future work could explore the system's adaptability to dynamic warehouse environments, investigate integration possibilities with other warehouse management systems, and consider energy efficiency and sustainability aspects of different pod selection and robot assignment strategies. These areas of future research have the potential to significantly refine and expand upon the foundations laid by this study, contributing to the ongoing evolution of efficient and adaptable robotic mobile fulfillment systems in e-commerce warehousing.

REFERENCES

1. Boysen, N., R. de Koster, and F. Weidinger, *Warehousing in the e-commerce era: A survey*. European Journal of Operational Research, 2019. **277**(2): p. 396-411.
2. de Koster, R., T. Le-Duc, and K.J. Roodbergen, *Design and control of warehouse order picking: A literature review*. European Journal of Operational Research, 2007. **182**(2): p. 481-501.
3. Boysen, N., D. Briskorn, and S. Emde, *Parts-to-picker based order processing in a rack-moving mobile robots environment*. European Journal of Operational Research, 2017. **262**(2): p. 550-562.
4. Kumar, S., J.-B. Sheu, and T. Kundu, *Planning a parts-to-picker order picking system with consideration of the impact of perceived workload*. Transportation Research Part E: Logistics and Transportation Review, 2023. **173**.
5. *Exclusive Photos from Amazon Warehouse Tour Bristol*. 2023; Available from: <https://www.aboutamazon.co.uk/news/operations/bristol-amazon-warehouse-tour>.
6. Chen, Y. and Y. Li *Storage Location Assignment for Improving Human–Robot Collaborative Order-Picking Efficiency in Robotic Mobile Fulfillment Systems*. Sustainability, 2024. **16**, DOI: 10.3390/su16051742.
7. Economist. *New Robots Smarter and Faster are Taking Over Warehouses* | Economist. 2022. [cited 2023 December 6]; Available from: <https://www.economist.com/science-and-technology/a-new-generation-of-smarter-and-faster-robots-are-taking-over-distribution-centres/21807595>.
8. Yang, X., et al., *Joint optimization of order sequencing and rack scheduling in the robotic mobile fulfillment system*. Computers & Operations Research, 2021. **135**.
9. Azadeh, K., R. De Koster, and D. Roy, *Robotized and Automated Warehouse Systems: Review and Recent Developments*. Transportation Science, 2019. **53**.
10. Nigam, S., et al., *Analysis of Class-based Storage Strategies for the Mobile Shelf-based Order Pick System*, in *13th IMHRC Proceedings (Cincinnati, Ohio. USA-2014)*. 2014. p. 18.
11. Lamballais, T., D. Roy, and M.B.M. De Koster, *Estimating performance in a Robotic Mobile Fulfillment System*. European Journal of Operational Research, 2017. **256**(3): p. 976-990.
12. Zou, B., et al., *Evaluating battery charging and swapping strategies in a robotic mobile fulfillment system*. European Journal of Operational Research, 2018. **267**(2): p. 733-753.
13. Yuan, R., T. Cezik, and S.C. Graves, *Velocity-Based Storage Assignment in Semi-Automated Storage Systems*. 2018: SSRN.
14. Zhuang, Y., et al., *Rack retrieval and repositioning optimization problem in robotic mobile fulfillment systems*. Transportation Research Part E: Logistics and Transportation Review, 2022. **167**.
15. Krenzler, R., L. Xie, and H. Li, *Deterministic Pod Repositioning Problem in Robotic Mobile*

- Fulfillment Systems*. ArXiv, 2018.
16. Xie, L., H. Li, and L. Luttmann, *Formulating and solving integrated order batching and routing in multi-depot AGV-assisted mixed-shelves warehouses*. European Journal of Operational Research, 2023. **307**(2): p. 713-730.
 17. Lu Zhen, Z.T., René de Koster, Shuaian Wang, *How to deploy robotic mobile fulfillment systems*. Transportation Science, 2023. **57**(6): p. 1403-1719.
 18. Xie, L., et al., *Introducing split orders and optimizing operational policies in robotic mobile fulfillment systems*. European Journal of Operational Research, 2021. **288**(1): p. 80-97.
 19. Zhuang, Y., et al., *Order picking optimization with rack-moving mobile robots and multiple workstations*. European Journal of Operational Research, 2022. **300**(2): p. 527-544.
 20. Zhang, S., et al., *Order picking optimization in a robotic mobile fulfillment system*. Expert Systems with Applications, 2022. **209**.
 21. Wang, B., X. Yang, and M. Qi, *Order and rack sequencing in a robotic mobile fulfillment system with multiple picking stations*. Flexible Services and Manufacturing Journal, 2022. **35**(2): p. 509-547.
 22. Teck, S. and R. Dewil, *Optimization models for scheduling operations in robotic mobile fulfillment systems*. Applied Mathematical Modelling, 2022. **111**: p. 270-287.
 23. Teck, S., P. Vansteenwegen, and R. Dewil, *An efficient multi-agent approach to order picking and robot scheduling in a robotic mobile fulfillment system*. Simulation Modelling Practice and Theory, 2023. **127**: p. 102789.
 24. Mojaver Tabrizi, A., et al., *A Three-Stage model for Clustering, Storage, and joint online order batching and picker routing Problems: Heuristic algorithms*. Computers & Industrial Engineering, 2023. **179**.
 25. Justkowiak, J.-E. and E. Pesch, *A column generation driven heuristic for order-scheduling and rack-sequencing in robotic mobile fulfillment systems*. Omega, 2023. **120**.
 26. Jiao, G., H. Li, and M. Huang, *Online joint optimization of pick order assignment and pick pod selection in robotic mobile fulfillment systems*. Computers & Industrial Engineering, 2023. **175**.
 27. Justkowiak, J.-E. and E. Pesch, *Stronger mixed-integer programming-formulations for order- and rack-sequencing in robotic mobile fulfillment systems*. European Journal of Operational Research, 2023. **305**(3): p. 1063-1078.
 28. Hashemi, S. and M. Ranjbar, *Optimal resource allocation and routing in robotic mobile fulfillment systems*. Computers & Operations Research, 2024. **165**: p. 106571.
 29. Lamballais Tessensohn, T., D. Roy, and R.B.M. De Koster, *Inventory allocation in robotic mobile fulfillment systems*. IIE Transactions, 2019. **52**(1): p. 1-17.

30. Jiang, W., et al., *Assignment of duplicate storage locations in distribution centres to minimise walking distance in order picking*. International Journal of Production Research, 2021. **59**(15): p. 4457-4471.
31. Merschformann, M., et al., *Decision rules for robotic mobile fulfillment systems*. Operations Research Perspectives, 2019. **6**.

APPENDIX

A.1 Basic Taguchi L9 Orthogonal Array

	Control factors and levels			
	A	B	C	D
1	1	1	1	1
2	1	2	2	2
3	1	3	3	3
4	2	1	2	3
5	2	2	3	1
6	2	3	1	2
7	3	1	3	2
8	3	2	1	3
9	3	3	2	1